GATE

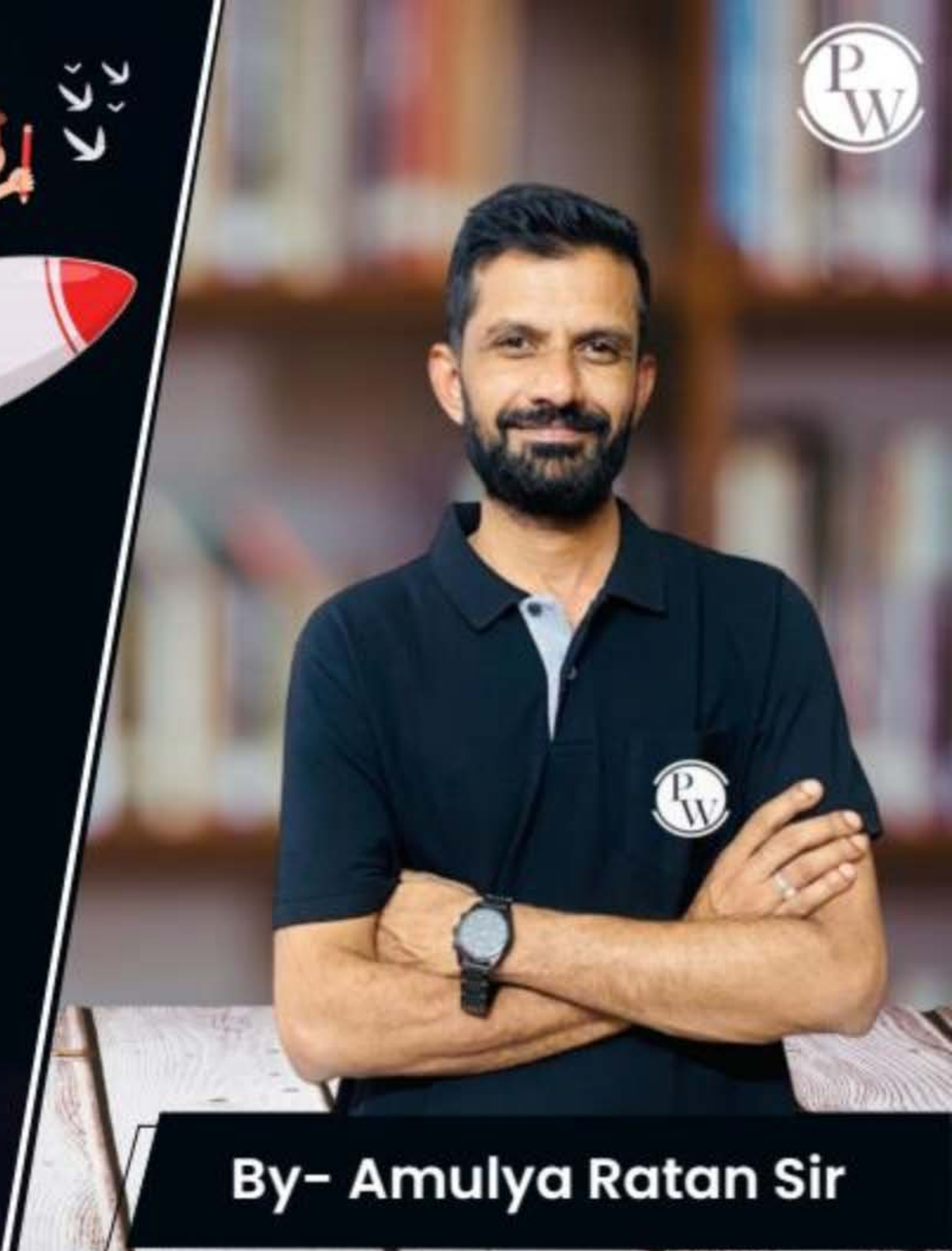
ALL BRANCHES



General Aptitude

QUANTITATIVE APTITUDE

Lecture No.- 15



By- Amulya Ratan Sir

Recap of Previous Lecture



Topic 1 Data Interpretation

Graph

Topics to be Covered



Topic-1 Mensuration 2-D

Topic-2 Mensuration 3-D

[MCQ]

S.P

#Q. If the income of the two companies were equal in 1996, what was the ratio of their expenditure?

C_A P/. ⇒ 35%

C_B P/. ⇒ 30%

$C_A = \frac{\text{Income}}{\text{Exp}} = 1.35$

$C_B = \frac{I}{\text{Exp}} = 1.3$

Assignment

$I_A = 1.35\ \text{Exp}$

$I_B = 1.3\ \text{Exp}$



(A) 26 : 27

(B) 100 : 67

(C) 1 : 2

(D) 5 : 27

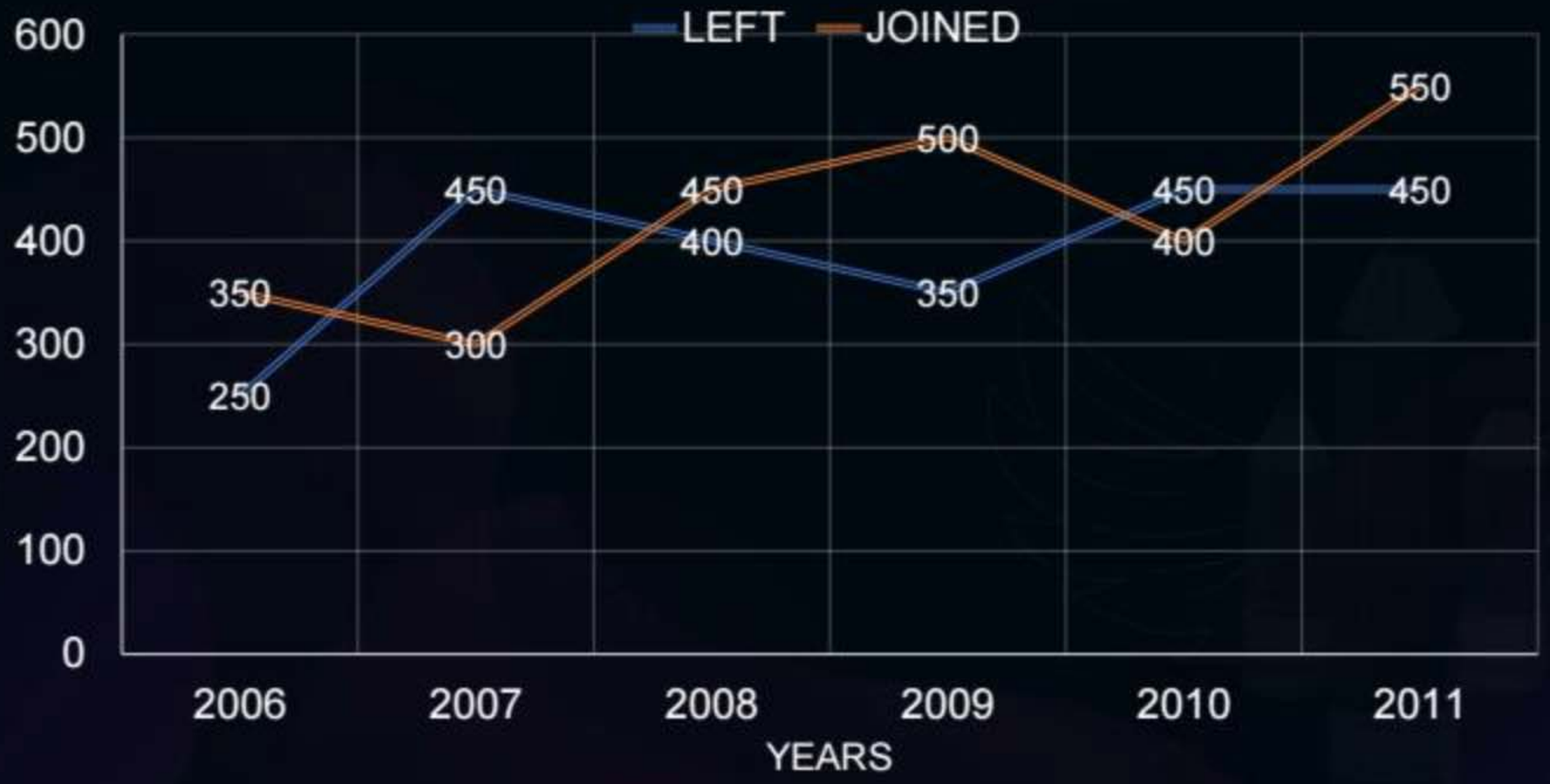
26:27

$1.35\ \text{Exp}_A = 1.3\ \text{Exp}_B$

$\frac{\text{Exp}_A}{\text{Exp}_B} = \frac{1.3}{1.35} = \frac{130}{135} \quad 26 \quad 27$

The following graph depicts the number of students who joined and left the school in the beginning of the academic year.

Strength of school in 2005 was 3000
2005 में स्कूल की संख्या 3000 थी



[MCQ]

#Q. By approximately what percent, the strength of school increased or decreased from 2007 to 2008?

Strength of school in 2005 was 3000



Assignment

A 1.2%

B 1.7%

C 2.1%

D 2.4%

$\frac{50}{2950} \times 100$

$= \frac{5000}{2950}$

$= 1.69$

$\approx 1.7\%$

[MCQ]

#Q. How many students were studying in the school during 2009?

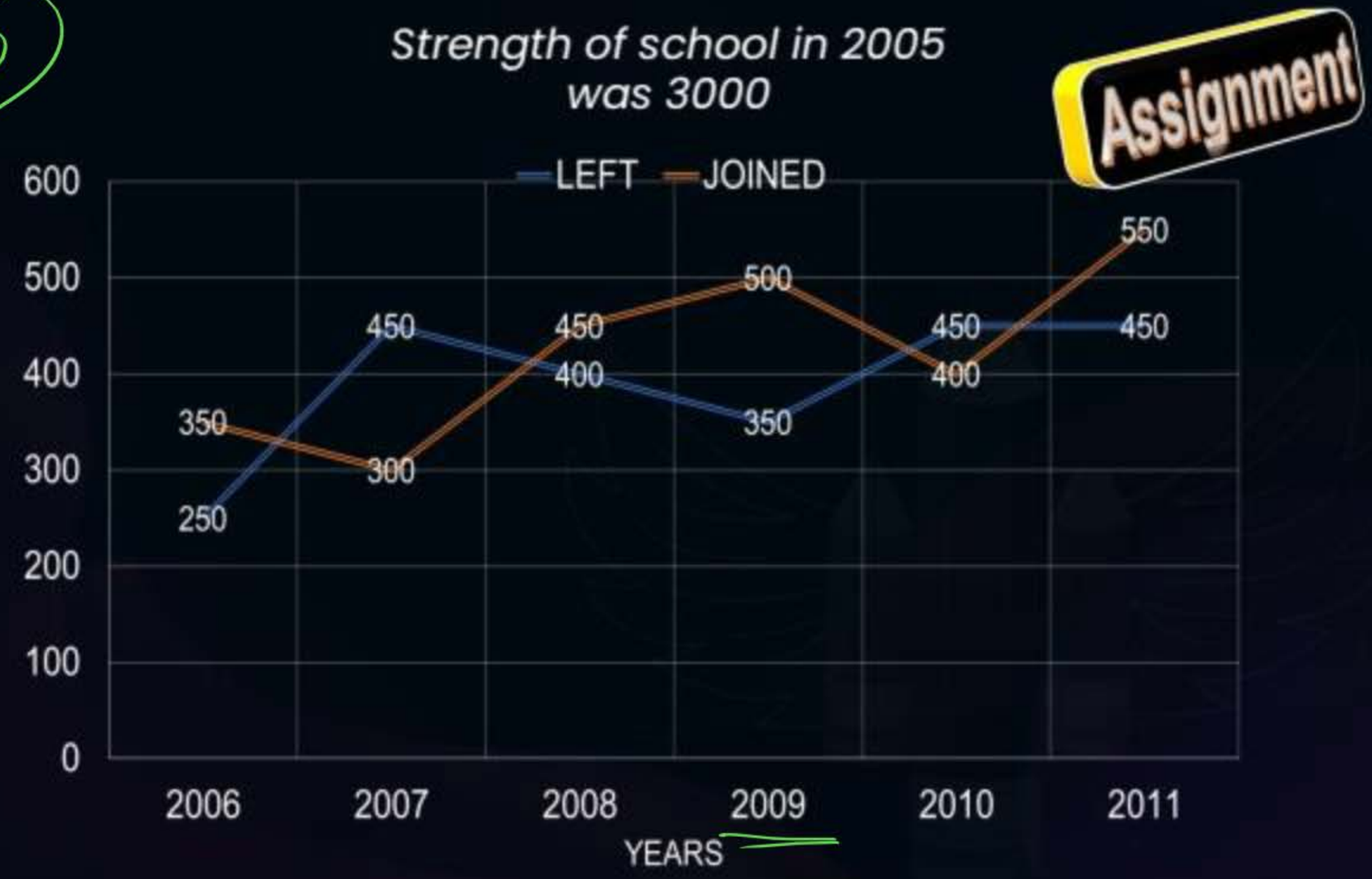
3150

A 2950

B 3000

C 3100

D 3150



Assignment

[MCQ]

#Q. The number of students studying in school in 2008 was what percent of the number of students studying in the scool in 2011?

$\frac{3000}{3200} \times 100$

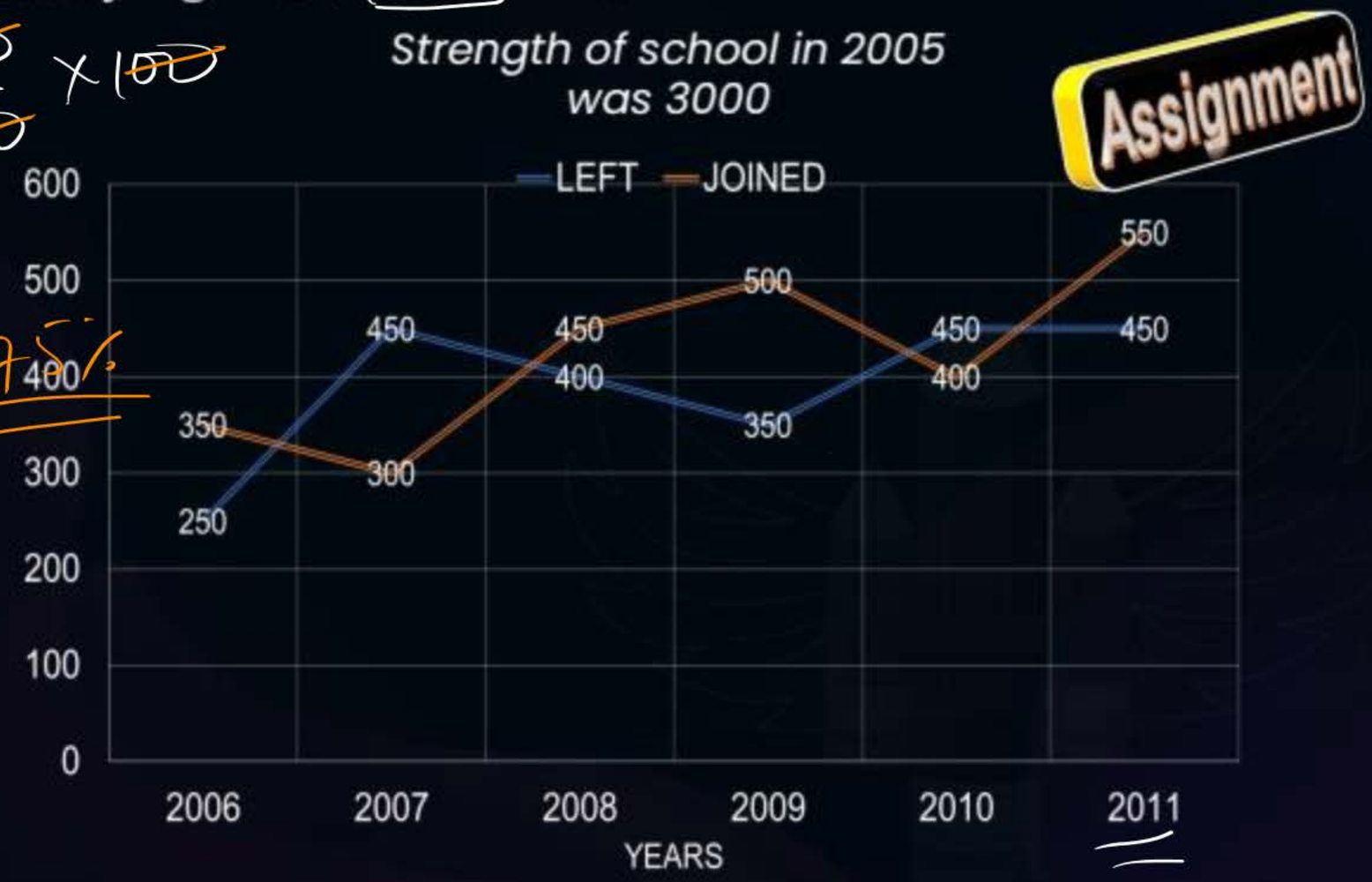
$= 93.75\%$

A. 92.13%

B. 93.75%

C. 97.25%

D. 96.88%



Assignment

[MCQ]

#Q. How many employee have their salary more than 3 times the income from bonus?

Source of Income	Employees				
	P	Q	R	S	T
Salary	400	200	700	300	400
Bonus	80	40	150	80	100
Overtime	180	70	200	170	200
Arrears	200	180	400	140	250
Misc	40	10	50	10	50
TOTAL	900	500	1500	700	1000



A One

B Five

C Three

D None

[MCQ]

#Q. Who among the following employees has maximum percentage of his salary out of the total income?

Assignment

Source of Income	Employees				
	P	Q	R	S	T
Salary	400	200	700	300	400
Bonus	80	40	150	80	100
Overtime	180	70	200	170	200
Arrears	200	180	400	140	250
Misc	40	10	50	10	50
TOTAL	900	500	1500	700	1000

P ⇒ $\frac{400}{9}\% = 44.4\%$

Q ⇒ 40%

R ⇒ $\frac{700}{15}\% = 46.6\%$

S ⇒ $\frac{300}{7}\% = 42.8\%$

T ⇒ 40%

A. T

B. P

C. S

D. R

[MCQ]

Arrear / Salary

#Q. Who among the following employees has minimum ratio of income from arrears to the income from salary?

Assignment

Source of Income	Employees				
	P	Q	R	S	T
Salary	400	200	700	300	400
Bonus	80	40	150	80	100
Overtime	180	70	200	170	200
Arrears	200	180	400	140	250
Misc	40	10	50	10	50
TOTAL	900	500	1500	700	1000

P ⇒ $\frac{200}{400} = 0.5$

Q ⇒ $\frac{18}{20} = \frac{9}{10} = 0.9$

R ⇒ $\frac{4}{7} = 0.57$

S ⇒ $\frac{14}{30} = \frac{7}{15} = 0.46$

T ⇒ $\frac{5}{8} = 0.625$

A. P

B. Q

C. T

D. S

MENSURATION: 2D

- Triangle
- Quadrilateral
- Circle
- Semicircle

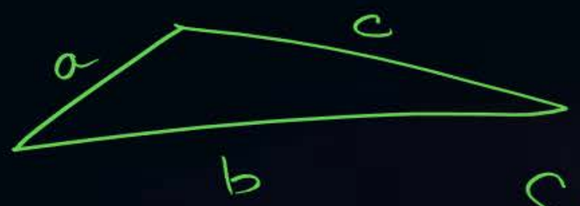


Poll

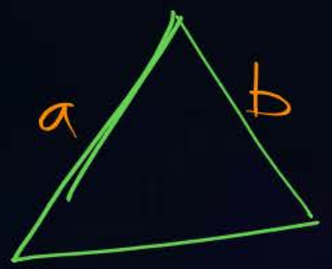
Triangles:

Types

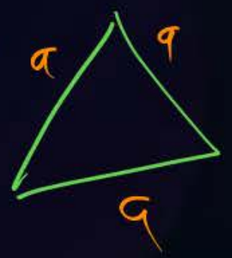
3 Types (side)



Scalene △



Isosceles △



Equilateral △

3 Types (Angle)

Acute ∠d △



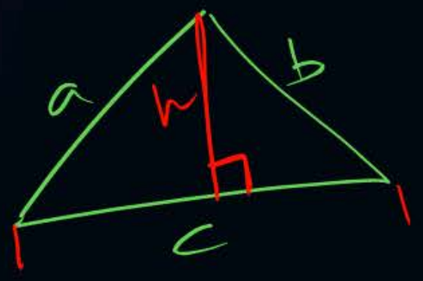
Right ∠d △



Obtuse ∠d △



Triangles:



Area

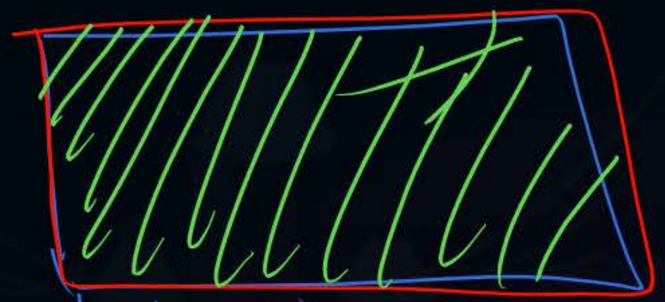
Perimeter

$P = a+b+c$

Heron's formula

$$\text{Area} = \sqrt{s(s-a)(s-b)(s-c)}$$

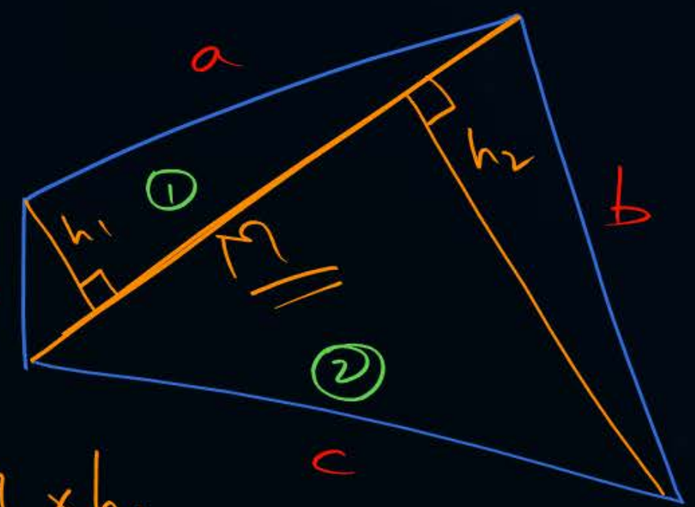
where $s = \frac{a+b+c}{2}$



$$\text{Area} = \frac{1}{2} \times \text{base} \times \text{height}$$

$$\text{Area Equilateral } \triangle = \frac{\sqrt{3}}{4} a^2$$

Quadilaterals:

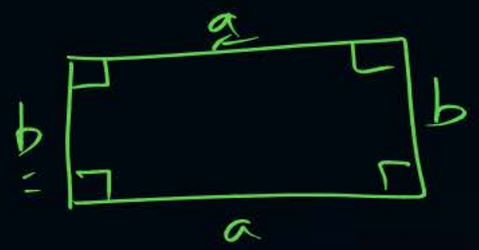


$$P = a+b+c+d$$

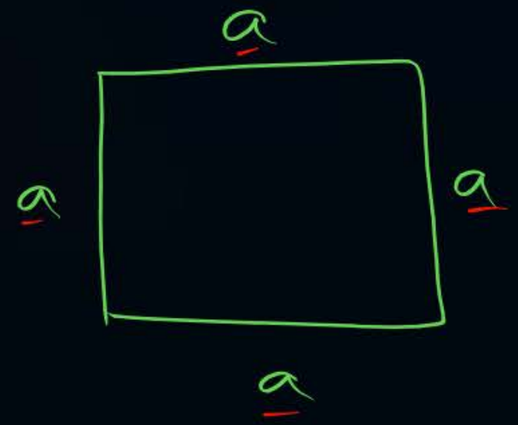
$$A = \frac{1}{2} \times m \times h_1 + \frac{1}{2} \times m \times h_2$$

$$= \frac{1}{2} \times \text{base} \times (h_1 + h_2)$$

Rectangle and Square:

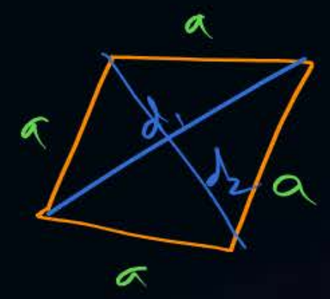


$$P = a+b+a+b$$
$$= 2(a+b)$$
$$A = l \times b$$
$$\therefore a \times b$$



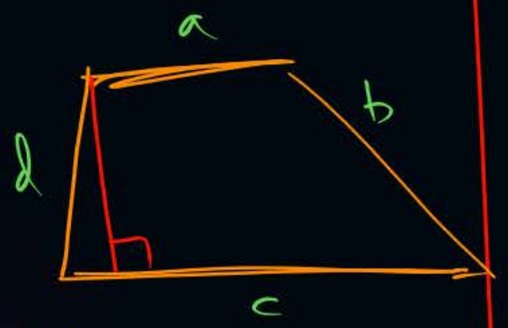
$$P = 4a$$
$$A = (a^2) \Rightarrow (\text{side})^2$$

Rhombus, Trapezium & Parallelogram:



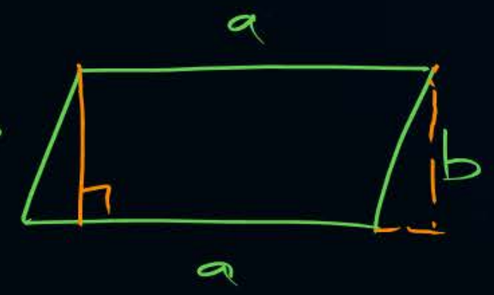
$$P = 4a$$

$$A = \frac{1}{2} \times d_1 \times d_2$$



$$P = a+b+c+d$$

$$A = \frac{1}{2} \times h \times (a+c)$$



$$P = 2(a+b)$$

$$A = \text{base} \times \text{height}$$

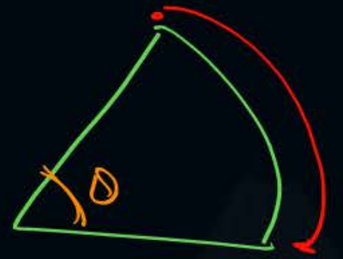
Circles & Sectors:

~~Perimeter~~ Circumference



Circumference $= 2\pi r$

Area $= \pi r^2$



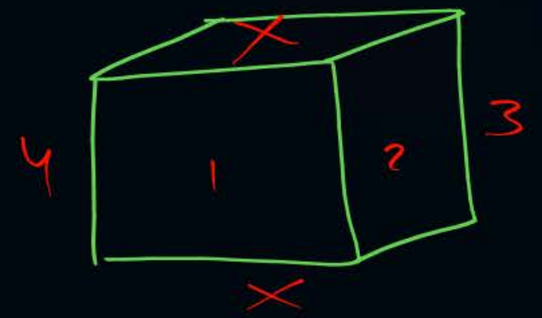
length of arc

Length of arc $= \frac{\theta}{360} \times 2\pi r$

Area of sector $= \frac{\theta}{360} \times \pi r^2$

MENSURATION: 3D

Volume

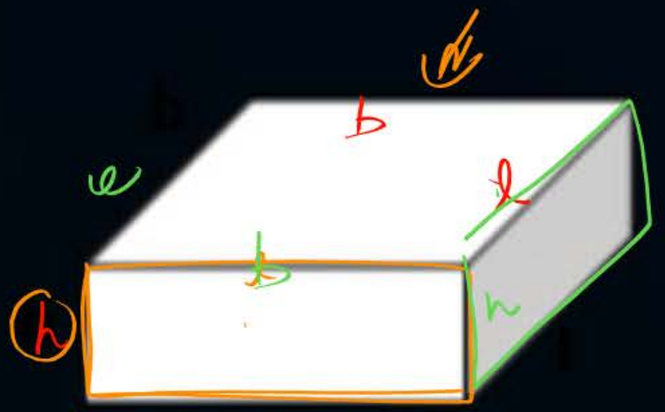


(LSA) Lateral Surface Area

(TSA) Total Surface Area

Cube
Cuboid
Cylinder
Cone
Sphere
Hemi-sphere

Cuboid:



$$\text{Volume} = l \times b \times h$$

$$\text{LSA} = 2bh + 2lh$$

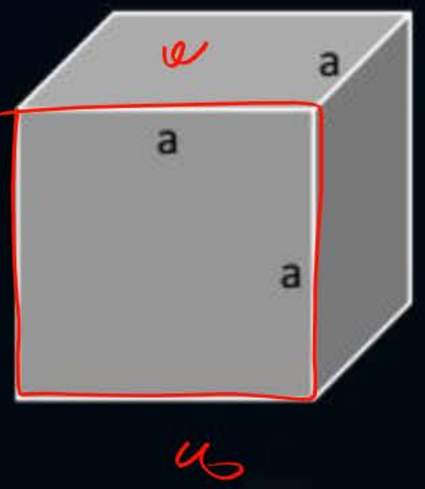
$$= 2h(l + b)$$

$$\text{TSA} = 2bh + 2lh + 2lb$$

$$= 2(lb + bh + hl)$$

Cube:

$V = a \times a \times a = (a)^3$



$LSA = 4a^2$

$TSA = 4a^2 + 2a^2 = 6a^2$

Cylinder:

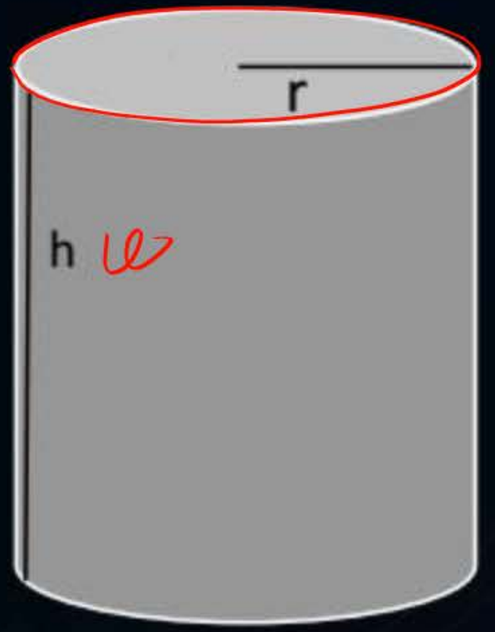
~~LSA~~ → Curved Surface Area (CSA)

$$V = \pi r^2 h$$

$$CSA = 2\pi r h$$

$$TSA = 2\pi r h + 2\pi r^2$$

$$= 2\pi r(h + r)$$



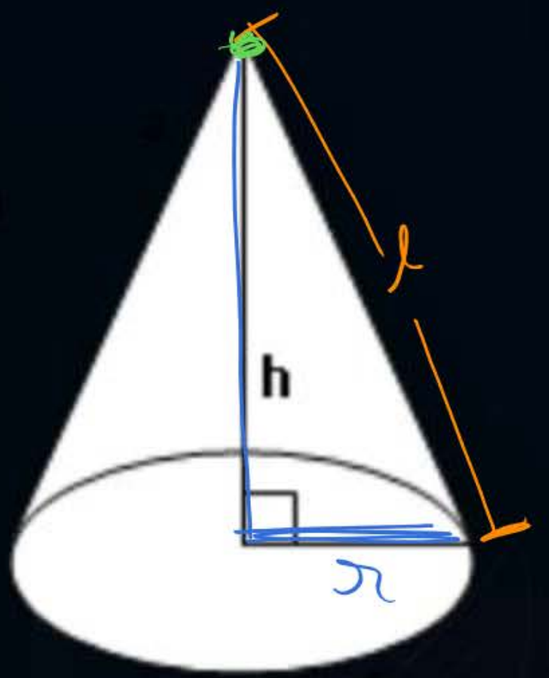
Cone:

$$V = \frac{1}{3}\pi r^2 h$$

$$CSA = \pi r l$$

$$TSA = \pi r l + \pi r^2$$

$$= \pi r(l + r)$$



Sphere:

$$V = \frac{4}{3}\pi r^3$$

$$TSA = 4\pi r^2$$



Hemi-sphere:

$$V = \frac{1}{2} \times \frac{4}{3}\pi r^3$$

$$= \frac{2}{3}\pi r^3$$



$$CSA = 2\pi r^2$$

$$TSA = 2\pi r^2 + \pi r^2 = 3\pi r^2$$

[MCQ]

#Q. What would be the area of sector of a circle whose radius is 12 cm and the length of the arc is 20 cm?

Area of Sector $= \frac{\theta}{360} \times \pi r^2$

Length of arc $= \frac{\theta}{360} \times 2\pi r$

$\Rightarrow 20 = \frac{\theta}{360} \times 2 \times \pi \times 12$ (×12)

$\therefore 120 = \frac{\theta}{360} \times \pi r^2$ (×r)

A. 60 sq. cm

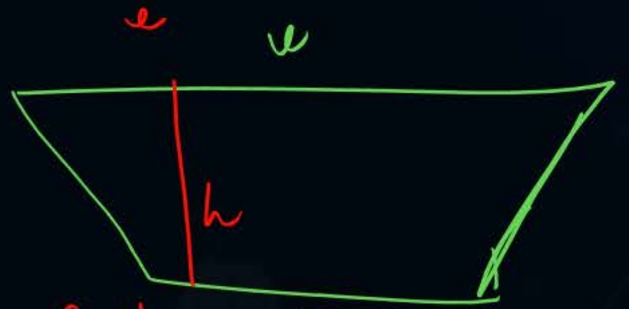
B. 240 sq. cm

C. 120 sq. cm

D. 64 sq. cm

[MCQ]

#Q. The cross-section of a canal is a trapezium in shape. If the canal is 7 m wide at the top and 9 m at the bottom and the area of cross-section is 128 sq m, find the height of the canal.



Area of Trapezium

$= \frac{1}{2} \times h \times$ (sum of 11 sides)

$\Rightarrow 128 = \frac{1}{2} \times h \times 16$

$\therefore h = \frac{128}{8} = 16$ m

A. 32 m

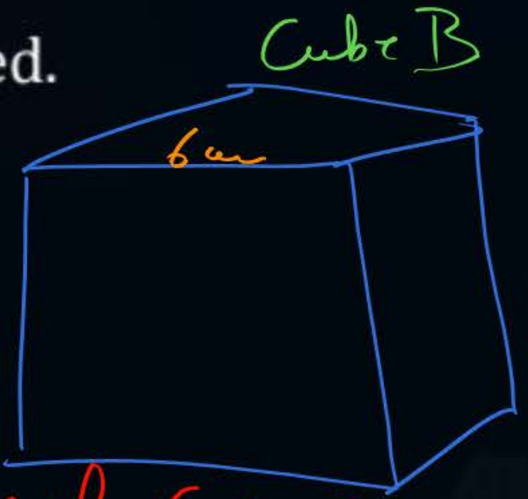
B. 8 m

C. 16 m

D. 4 m

[MCQ]

#Q. A cube of 6 cm side melted and smaller cubes of 2 cm side are manufactured. Find the number of smaller cubes so formed.





Volume of Cube B

$= n \times$ Volume of Cubes

$\Rightarrow 6^3 = n \times 2^3$

$\Rightarrow \frac{216}{8} = n = 27$

A 12

B 27

C 24

D 8

[MCQ]

#Q. The length, breadth & height of a cuboid are in the ratio of 4 : 3 : 2 and its volume is 3000 m^3. Find its total surface area.

Assignment

A $1300\ m^2$

B $1500\ m^2$

C $1333\ m^2$

D $2700\ m^2$

[MCQ]

#Q. The dimensions of a room are 20 m x 15 m x 10 m. What is the cost of painting its four walls at the rate of ₹ 15 per 100 sq m?

Assignment

A ₹ 70

B ₹ 105

C ₹ 225

D None of these

[MCQ]

#Q. In two concentric circles, a chord length 80 cm of larger circle becomes a tangent to the smaller circle whose radius is 9 cm. The radius of the larger circle will be?

Assignment

A 13 cm

B 41 cm

C 52 cm

D 75 cm

[MCQ]

#Q. The diagonal of a rectangle is $\sqrt{41}$ cm and its area is 20 sq cm. The perimeter of the rectangle is?

Assignment

A 9 cm

B 18 cm

C 20 cm

D 41 cm



2 mins Summary

Topic Mensuration 2D & 3D

THANK - YOU